LeepTalent Level 3 End-point Assessment

Learner Guide

UFO Sightings Investigation — what to do, why, and where you are in the process | ★★★★☆

This guide shows one sound way to complete the scenario, the reasoning behind each step, and which stage of the data analysis process each step belongs to. This is a cleaning-heavy scenario — most of the marks are won by finding and fixing the data-quality issues before you analyse. Work the task yourself first, then check your method here. Screenshots are illustrative mock-ups.

The data analysis process — follow it in order:

Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report

Each step is tagged with its stage. The Clean/Prepare stage is the heart of this scenario.

Scenario 1 — Data Gathering and Validation

Step 1 — Identify and import

STAGE 1 · IDENTIFY & 2 · COLLECT

What you do: Import both CSVs, save as **XFiles_Analysis.xlsx**, and confirm both tables share an **AgentID** field. Note the known issues up front: messy text, a duplicate agent, mixed date formats and a sighting whose agent may not exist.

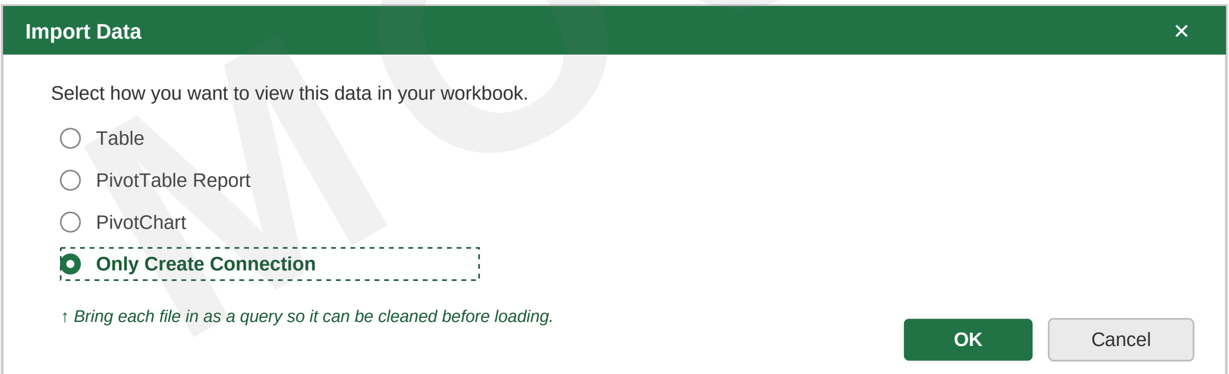


Figure 1 — Bring each file in as a query so it can be cleaned before loading.

> **Why scope the mess first?** Knowing what kinds of problems to expect means you clean deliberately and don't miss anything — which is exactly what the validation part of this scenario tests.

Step 2 — Inspect for problems

STAGE 3 · CLEAN / PREPARE

What you do: Sort and filter each column to surface inconsistencies — mixed capitalisation, extra spaces, blank rows, duplicate Agent IDs and missing values.

> **Why inspect before cleaning?** Sorting a column groups similar values together, so "london" and "London", or a stray duplicate ID, become obvious. You can't fix what you haven't found.

Step 3 — Clean the text data

STAGE 3 · CLEAN / PREPARE

What you do: Tidy the text with the cleaning functions, remove duplicates, and reconcile inconsistent region names.

1. `=TRIM(A2)` — strip leading, trailing and double spaces.
2. `=PROPER(TRIM(A2))` — consistent capitalisation for names and locations.
3. `=UPPER(...)` — consistent codes/status where needed.
4. **Data → Remove Duplicates** — AgentID **A020** appears twice.
5. Fix the mis-spelled **"Scotand" → "Scotland"** and the double-spaced **"East Of England"** so regions group correctly.

Cleaning the agent data — TRIM, PROPER, UPPER + Remove Duplicates

Before (raw, messy):

AgentName	Region	Status
" Dana scully "	"south east "	" Active"
FOX MULDER	North east	Active
Monica reyes	north West	Active

`=TRIM(A2)` *removes extra spaces*

`=PROPER(TRIM(A2))` *tidies case*

After (cleaned):

AgentName	Region	Status
Dana Scully	South East	Active
Fox Mulder	North East	Active
Monica Reyes	North West	Active

`=UPPER(C2)` *for codes/status*

⚠ Also: AgentID A020 appears twice → use Data → Remove Duplicates. And fix "Scotand" → "Scotland".

Figure 2 — TRIM/PROPER/UPPER tidy the messy text; Remove Duplicates handles the repeated AgentID.

> **Why clean before blending or analysing?** Untrimmed or mis-cased values break both lookups and grouping — " Dana scully " won't match "Dana Scully", and "Scotand" will appear as a separate region in your pivot, fragmenting the results. Cleaning first makes everything downstream reliable.

Step 4 — Blend, and check every match

STAGE 3 · CLEAN / PREPARE

What you do: Use XLOOKUP to bring Region and ExperienceYears into the sightings table on AgentID, then check that *every* lookup returned a value:

```
=XLOOKUP([@AgentID], Agents[AgentID], Agents[Region])
```

Blend with XLOOKUP — and check every match returns a value

SightingID	AgentID	Region (looked up)	ExperienceYears
S019	A018	Wales	5
S020	A020	Scotland	15
S021	**A099**	**#N/A**	**#N/A**

⚠ A099 does not exist in AgentProfiles — the lookup returns #N/A.

Don't hide it with IFERROR and move on — report it: this sighting has no matching agent record.

Figure 3 — One sighting (A099) has no matching agent — the lookup returns #N/A. Report it, don't hide it.

> **Why check for failed matches?** Sighting **S021** is logged against agent **A099**, which doesn't exist in the profiles, so its lookup returns #N/A. The brief asks you to say why a lookup fails — the right answer is to flag that this sighting has no matching agent record, not to bury it with IFERROR. Spotting and explaining it is part of the assessment.

Scenario 2 — Data Analysis

Step 5 — Calculated columns

STAGE 4 · ANALYSE

What you do: Add two columns:

`Classification: =IF([@CredibilityScore]>=80,"High Credibility","Medium Credibility")`

`Adjusted Duration: =[@DurationMins]/[@ExperienceYears]` (format to 2 dp)

> **Why these columns?** Classification turns a raw score into a band you can group and filter by; Adjusted Duration normalises time by experience so a long sighting by a junior agent is comparable to a senior one. Both are derived by formula so they update if the data changes.

Expected result: with the 80+ threshold, **13 of 22** sightings are "High Credibility" and 9 are "Medium". Overall average credibility is about **79.5**.

Step 6 — Conditional formatting

STAGE 5 · VISUALISE

What you do: Highlight Credibility Score > 90 in green and Duration > 120 minutes in orange.

> **Why conditional formatting?** It makes the standout cases jump off the page without re-reading every row — the most credible and the most time-intensive sightings become visible at a glance.

Step 7 — Regional insights (SUMIF / AVERAGEIF)

STAGE 4 · ANALYSE

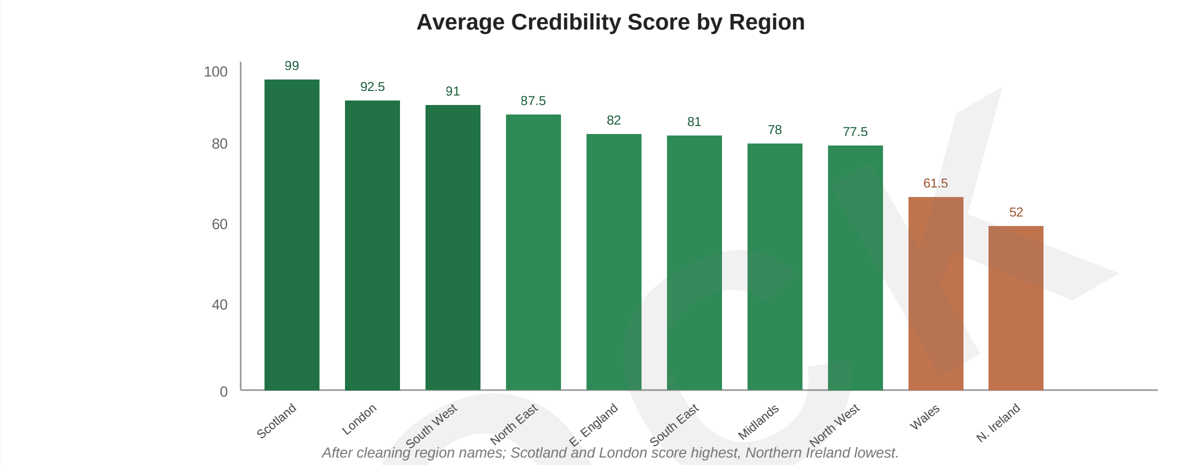
What you do: `=AVERAGEIF(Region range,"Scotland",Score range)` for average credibility per region, and SUMIF for total duration.

> **Why these only work after cleaning:** if "Scotand" and "Scotland" both exist, or regions carry stray spaces, AVERAGEIF treats them as different regions and your averages fragment. This is why Step 3 matters before Step 7.

Step 8 — Pivot Table and Chart

STAGE 5 · VISUALISE

What you do: Pivot Table with Region as rows, and Average Credibility plus Count of Sightings as values. Add a column Pivot Chart named **Regional_Credibility_Chart**.



After cleaning region names; Scotland and London score highest, Northern Ireland lowest.

Figure 4 — Average credibility by region (after cleaning region names).

> **Expected result:** Scotland and London report the most credible sightings (~99 and ~92.5), while Northern Ireland (~52) and Wales (~61.5) are lowest.

Step 9 — Interpret your findings

STAGE 6 · INTERPRET / REPORT

What you do: Write a short summary: the most credible regions, and whether experience relates to credibility.

> **On experience vs credibility:** comparing agents' ExperienceYears against their sightings' credibility shows a moderate positive relationship (correlation about 0.6) — more experienced agents do tend to file more credible reports, which is a worthwhile, defensible finding to report.

"After cleaning (trimming spaces, standardising case, removing a duplicate agent and fixing the 'Scotand' mis-spelling), the data shows Scotland and London reporting the most credible sightings (~99 and ~93), with Northern Ireland and Wales lowest. More experienced agents tend to file more credible reports (correlation ≈ 0.6). One sighting (S021) is logged against a non-existent agent (A099) and should be queried with the field team."

Marker's checklist — process followed in order

- **Identify/Collect:** both CSVs imported, workbook saved, AgentID confirmed as the shared key.
- **Clean/Prepare:** TRIM/PROPER/UPPER applied; duplicate A020 removed; "Scotand" and double-spaced regions reconciled; dates standardised; failed lookup (A099/S021) found and explained.
- **Analyse:** Classification (IF, ≥80) and Adjusted Duration columns; AVERAGEIF/SUMIF regional insights.
- **Visualise:** conditional formatting; Pivot Table and Regional_Credibility_Chart.
- **Interpret/Report:** summary incl. most-credible regions and the experience-vs-credibility relationship.